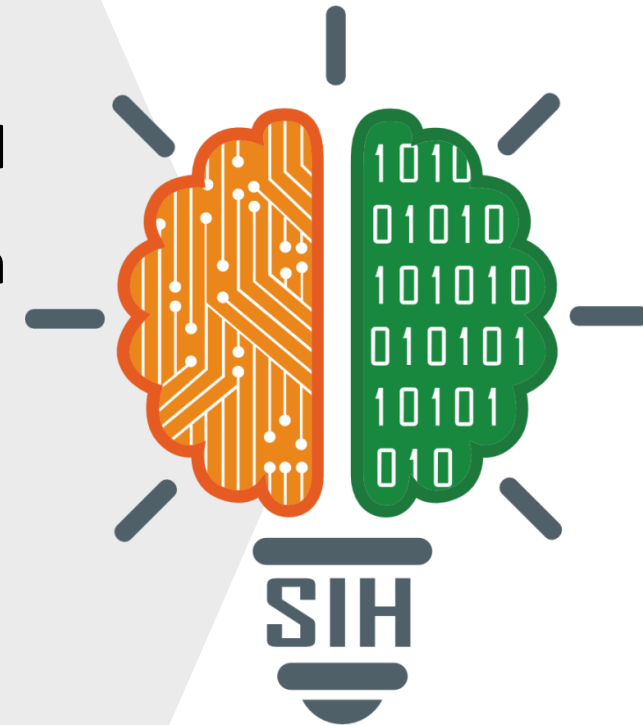


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID –SIH26003**
- **Problem Statement Title- AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID-1**
- **Team Name (Registered on portal)- Sidequest**



SAHARA – Smart Dementia Support & Cognitive Gaming Platform

Proposed Solution

SAHARA is a dementia-support application with separate **Patient** and **Caretaker** interfaces.

Patient: 3 cognitive games — **Memory Match**, **Sequence Recall** & **Family Recognition**.

Caretaker: Monitors **activity**, **scores**, **schedules**, **medicine status** & **family photos**.

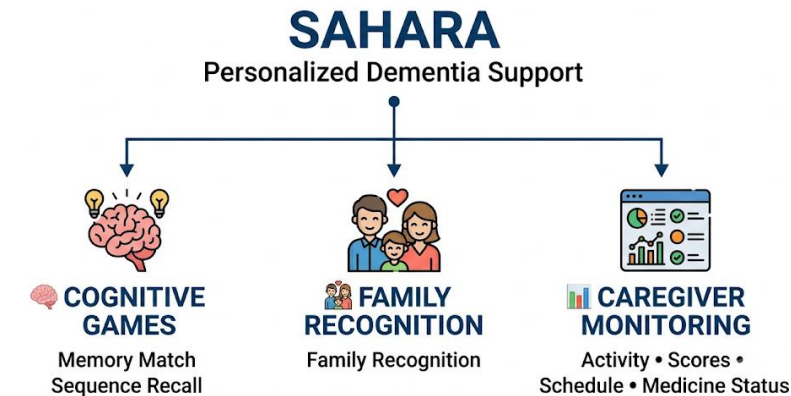
Accessibility: **Voice-assisted instructions** and language support.

How it addresses the problem

Provides **simple, engaging and personalized cognitive activities** while helping caregivers monitor patient activity and performance.

Innovation

Combines **cognitive gaming + family recognition + caregiver monitoring**, with **randomized content and accessibility features** in one platform.



Technologies used

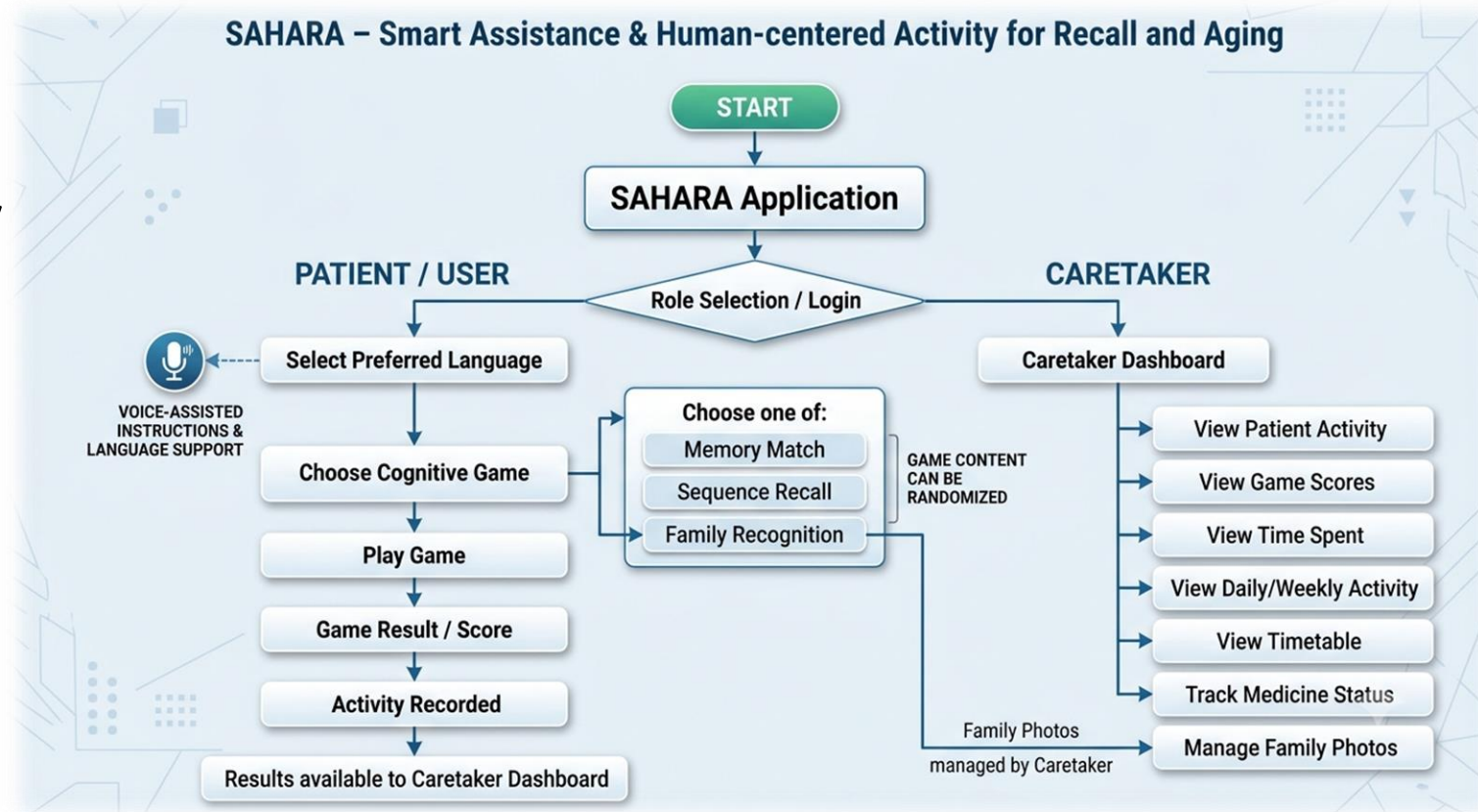
HTML + CSS + TypeScript → Frontend

Local Storage / Database → Data

Speech Technology → Voice

Role-Based Authentication → Security

Process Flowchart



FEASIBILITY



- **Technical Feasibility:** HTML + CSS + TypeScript support the cognitive games, dashboard and responsive interface.



- **Operational Feasibility:** Simple, elderly-friendly interaction with large buttons and minimal navigation.



- **Scalability:** Prototype can transition from local storage to a secure backend/database.



- **Accessibility:** Voice assistance and multilingual support improve accessibility.

CHALLENGES



- **Device & Browser Compatibility:** Voice features may not work consistently across all devices and browsers.



- **Internet Dependency:** Online-dependent services may have limited functionality without connectivity.



- **Speech Recognition Limitations:** Accuracy can vary with language, accent, background noise and device capability.



- **Data & Storage Scalability:** The prototype uses local storage; production deployment requires secure backend infrastructure.

STRATEGIES



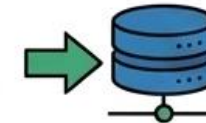
- **Compatibility Testing:** Test voice and application features across different devices and browsers.



- **Offline-Friendly Design:** Keep core game functionality usable wherever possible without continuous connectivity.

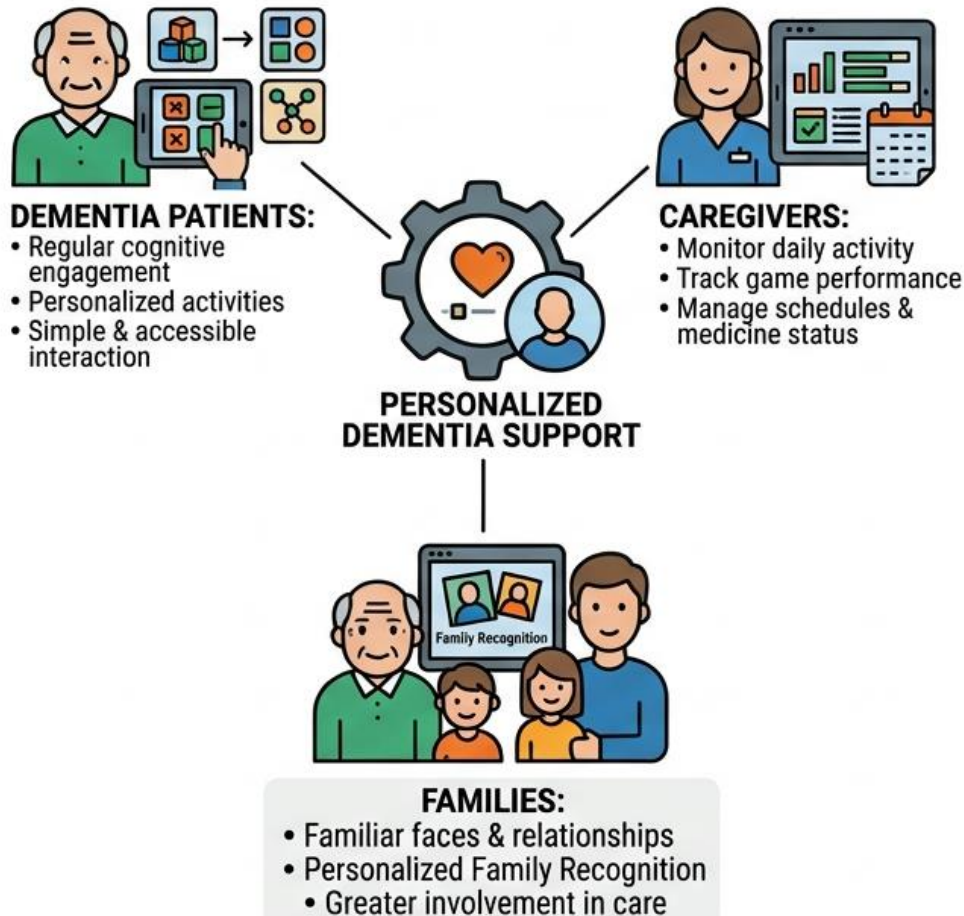


- **Speech Optimization:** Use clear speech, supported languages and suitable device/browser capabilities.

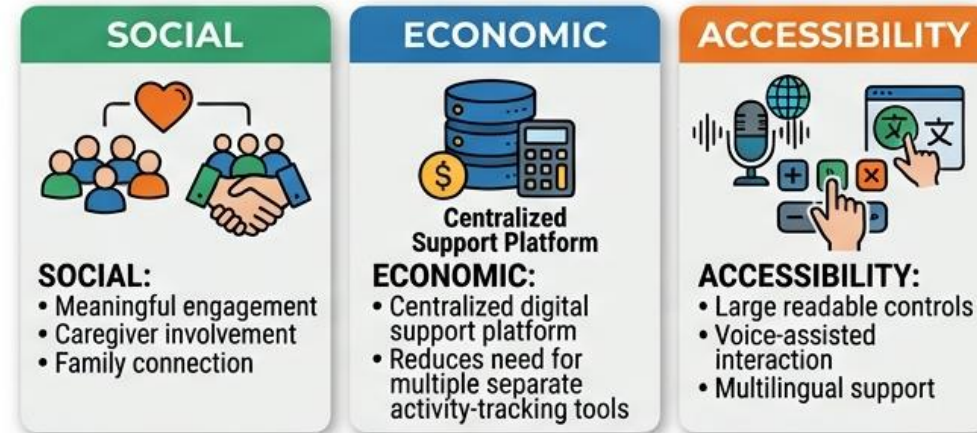


- **Secure Backend Migration:** Move prototype data to secure persistent storage for production and multi-user deployment.

POTENTIAL IMPACT ON TARGET AUDIENCE



BENEFITS OF THE SOLUTION



OVERALL IMPACT



- **Details / Links of the reference and research work**
- [Effectiveness of Internet-Based or Mobile App Interventions for Family Caregivers of Older Adults with Dementia: A Systematic Review – PubMed](#) : Internet and mobile-based interventions can improve the well-being and support needs of dementia caregivers.
- [Home-Based Digital Healthcare Interventions for Dementia: A Systematic Review of Patient and Family Caregiver Outcomes - PubMed](#) : Digital, personalized, home-based interventions can support cognitive health in dementia while reducing caregiver burden and stress.